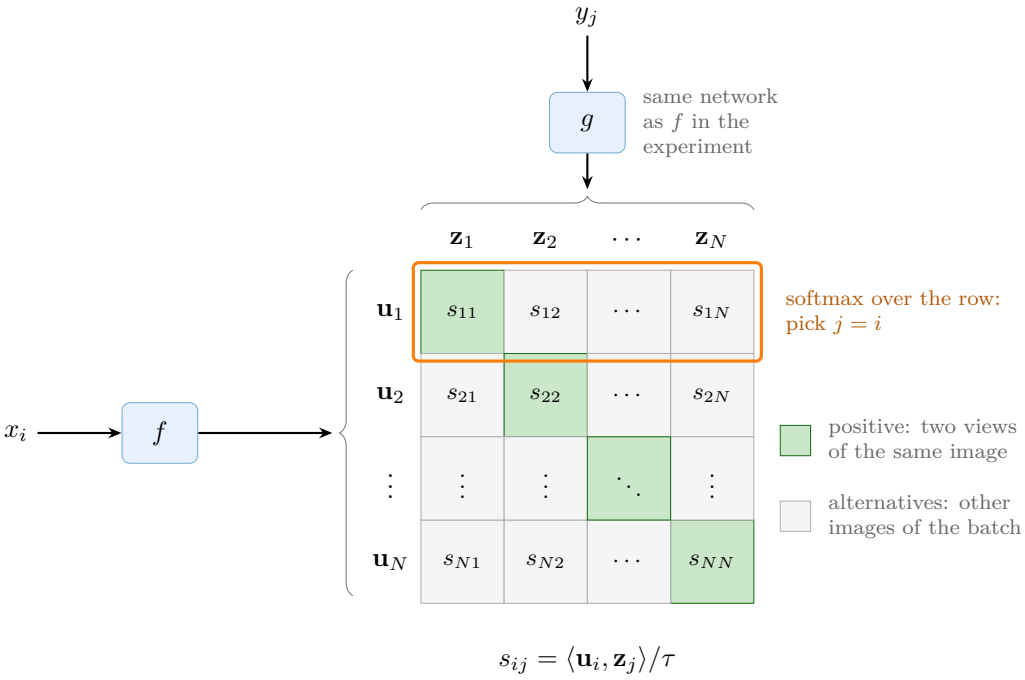

y_j
g
same network as f in the experiment
$\mathbf{z}_1$ $\mathbf{z}_2$ $\cdots$ $\mathbf{z}_N$
$\mathbf{u}_1$ s_{11} s_{12} $\cdots$ s_{1N}
$\mathbf{u}_2$ s_{21} s_{22} $\cdots$ s_{2N}
$\mathbf{u}_N$ s_{N1} s_{N2} $\cdots$ s_{NN}
softmax over the row: pick $j = i$
positive: two views of the same image
alternatives: other images of the batch
x_i
f
$s_{ij} = \langle \mathbf{u}_i, \mathbf{z}_j \rangle / \tau$